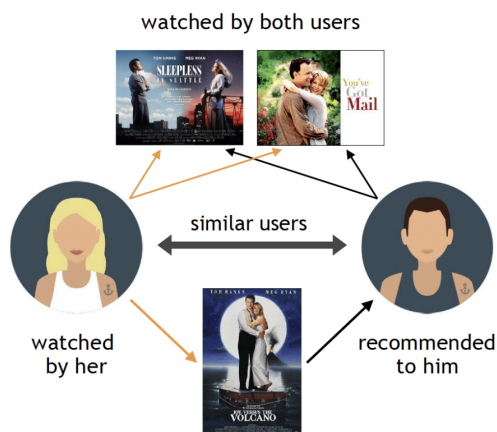


Easy website

- Suppose you're listening to a song on Spotify, watching a video on YouTube or Netflix, or shopping on Amazon; you'll always see a list of similar songs, videos, or products recommended to you. Amazingly, most of them perfectly match your tastes
- Recommender systems are designed to understand users' and products' preferences, past decisions, and characteristics by analyzing interaction data such as impressions, clicks, likes, and purchases.
- Three main types of recommendation systems are widely used:
 - Content-Based - recommends items by matching the attributes of items with users' preferences.
 - Collaborative Filtering
 - Context filtering
- These recommendation systems rely on feedback i.e. implicit & explicit
- This feedback is abundant and easy to collect, but it is less precise and can be noisy.
- Each item in the system is described by a profile that captures its essential characteristics.
- Term Frequency (TF): This is a count of how often a particular word appears in a document relative to the most frequent word in that document.
- Inverse Document Frequency (IDF): This calculates how unique a word is across all documents; more unique words are given higher importance.
- For content based recommendation, we use two main methods:
- Cosine Similarity – This method calculates the cosine of the angle between the item profile vector and the user profile vector. A smaller angle (or higher cosine similarity) suggests a more robust match between the user's preferences and the item's characteristics.
- Classification Models – Methods like decision trees can be used to decide whether an item would appeal to a user based on various characteristics of the user and the item.

Collaborative Filtering



- CF focuses on user-item interactions, utilizing the feedback or activity history of all users to make predictions.

- predicts a user's preferences by using the collective knowledge and behaviors of a large pool of users.
- Memory-Based CF – Uses the entire user-item interaction matrix to make direct recommendations based on similarities between users or items. It is straightforward but can struggle with large, sparse matrices. Generally, it deals with implicit feedback. It consists of two methods: user-based collaborative filtering and item-based collaborative filtering.
- Model-Based CF – Uses machine learning models to predict interactions between users and items. Techniques like matrix factorization, clustering, SVD, and deep learning are used to learn latent features from the data, improving prediction accuracy. We also use these methods for explicit feedback.
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